

Power and design analysis

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Agenda

1. Statistical error rates and power
2. Design analysis by simulation
3. A quantitative genetic study design
4. Controlling genetic effect sizes

Statistical error rates and power

We should spend more time and brain power designing studies that accurately measure what they are supposed to measure.

Is the sample size big enough? Is the statistical model appropriate? Can the parameter be measured? Is this assumption okay?

What do we expect to see if there is an effect?
If there isn't one?

Samples and populations

Population: The set of items (e.g., individuals, animals etc) that we want information about

Sample: The subset of the population we observe.

Note: “sample” often used to refer to a single unit (e.g., a blood sample from one individual); here it refers to a set of individuals sampled.

Statistics and parameters

We make inferences about the population based on the sample.

Population parameter: number that describes the population

Example: population mean μ

Sample statistic: a number calculated from the sample

Example: sample mean $\bar{y}$

Statistical model

We assume that the observations are drawn from a random distribution, in this case a normal distribution with some mean and standard deviation.

$$X \sim N(\mu, \sigma)$$

Statistical models are stochastic. This means that we can **simulate**.

Statistical inference from the sampling distribution

The probability distribution of a sample statistic

Describes how the outcome would vary if we repeated the sampling

Helps quantify the precision of estimates (assuming the model is good)

Hypothesis testing

Null hypothesis: H_0 : A default hypothesis of no difference

Alternative hypothesis H_1 : A hypothesis contrasted against the null

Reject null hypothesis when

- Test statistic greater than critical value (e.g., t-value of 2.26)
- p-value less than threshold (e.g., $p < 0.05$)
- Confidence interval that doesn't overlap zero

Example: two groups of chickens

BW35 (g)

Group A: 1769 1977 1755 2264 1831 1616 1632 2118 1914 2067

Group B: 1938 2308 2134 2475 1845 2279 2272 2043 1950 2318

We want to know: Are they different?

Output from R

```
Welch Two Sample t-test
data:  group1 and group2
t = -2.798, df = 17.975, p-value = 0.0119
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -458.60144  -65.23182
sample estimates:
mean of x mean of y
 1894.382  2156.299
```

hypothesis

p-value < 0.05

95% confidence interval that doesn't include 0

Type 1 error: **erroneously reject the null** when it is true.

The type 1 error rate is called α , used to set significance thresholds etc.

Type 2 error: **fail to reject the null** when it is false.

Power: probability of correctly rejecting the null hypothesis, for a given effect size and study design.

$1 - \beta$, where β is the type 2 error rate.

Type S (sign) error rate: probability that a statistically significant estimate has the wrong sign.

Type M (magnitude) error ratio: expectation of absolute value of a statistically significant estimate divided by true effect size.

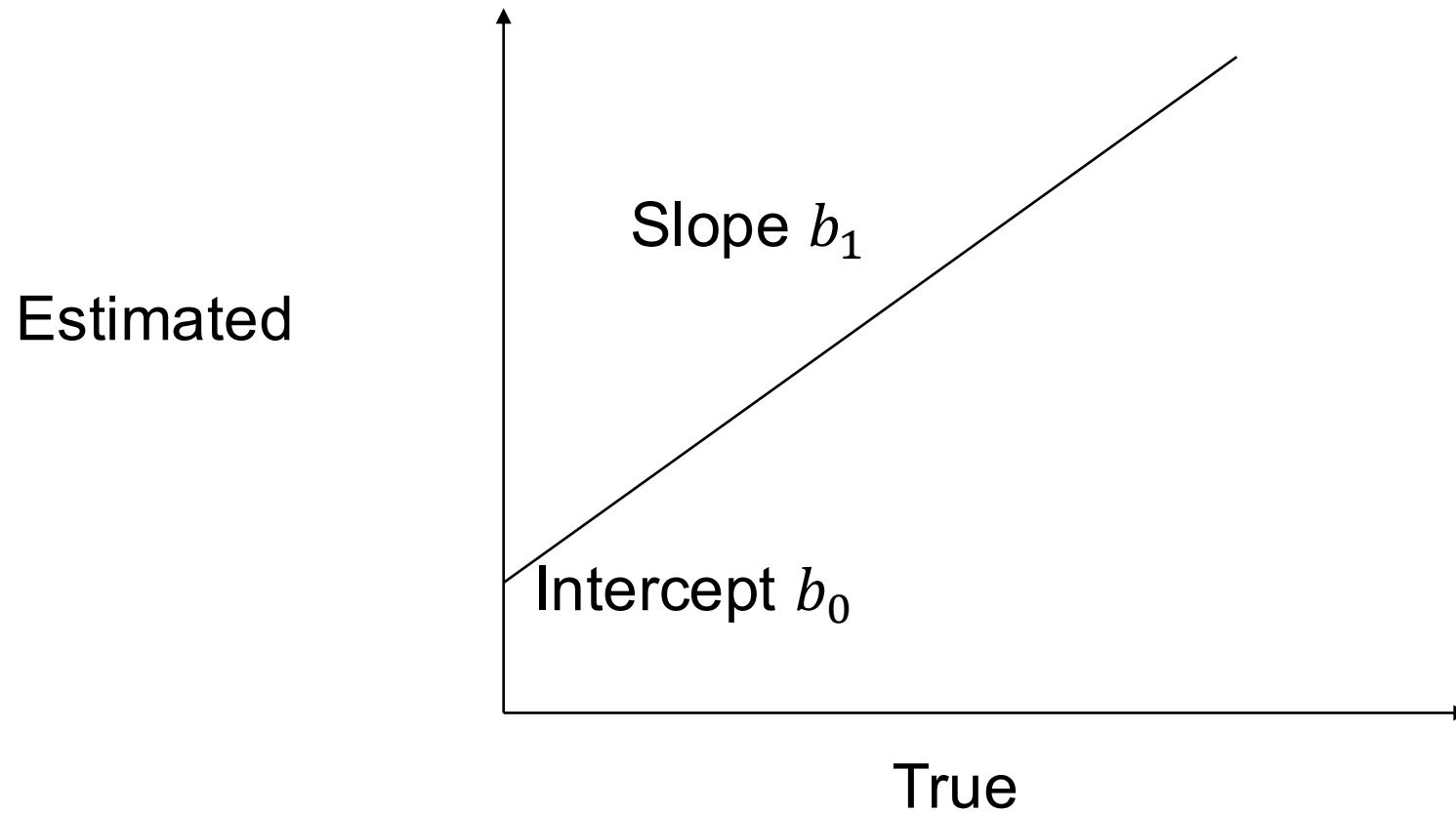
(Gelman & Carlin 2014)

Crucial insight: **underpowered** studies don't just risk missing true effects, they also **overestimate effects** (sometimes dramatically).

Probably relates to "decline effect", "replication crisis", and other concerns about the validity of science.

Bias: systematic error in estimation

In animal breeding literature: offset b_0 and dispersion b_1 .



In classification and in genomics:

False positive rate: type 1 error rate

False discovery rate (FDR): fraction significant estimates that are false positives

And many many many more ways to measure accuracy and precision.

When you **know the true result**, you can calculate anything!



Design analysis by simulation

Sometimes there are formulas for power (e.g, the t-test).

For more complex situations or when we can't be bothered, we can always simulate.

Assume we have two groups of chickens, measuring weight at day 35. What sample size do we need to reliably detect the effect of an intervention that promotes growth?

Assume: Mean 1840 g, standard deviation 191 g. Effect size 100 g.

```
a <- 1840
```

```
b <- 100
```

```
sd <- 191
```

```
d <- b / sd
```

```
simulate_t <- function(a, b, sd, n_group) {  
  group <- rep(0:1, each = n_group)  
  trait <- a + group * b + rnorm(n_group * 2, 0, sd)  
  broom::tidy(t.test(trait ~ group))  
}
```

Assume: Mean 1840 g, standard deviation 191 g. Effect size 100 g.

`d <- b / sd`

Cohen's d: standardised difference between groups

$$d = \frac{b}{\sigma} = \frac{100}{191} \approx 0.52$$

Rules of thumb, $d = 0.5$ considered "moderate" effect size

Assume we sample 60 chickens per group.

We run the simulation 1000 times and estimate a power of 0.8,
Type S error rate of 0, Type M ratio of 1.1.

```
sim_test <- purrr::map_dfr(1:1000, function(x) simulate_t(a, b, sd, 60))
```

```
est_power <- sum(sim_test$p.value < 0.05)/nrow(sim_test)
```

```
est_S_error <- sum(sim_test$p.value < 0.05 & sim_test$estimate > 0)/nrow(sim_test)
```

```
est_M_error <- mean(abs(sim_test$estimate[sim_test$p.value < 0.05])/b)
```

Using the power formula for the t-test in R:

```
power.t.test(delta = b, sd = sd, sig.level = 0.05, power = 0.8)
```

Two-sample t test power calculation

```
      n = 58.24311
  delta = 100
     sd = 191
sig.level = 0.05
  power = 0.8
alternative = two.sided
```

NOTE: n is number in *each* group

A quantitative genetic study design

Case study: **genetic correlation** between two environments.

Siblings are put in two different environments.

Genetic correlation as an estimate of gene-by-environment interaction.

Reading: "Simulating quantitative genetic data to analyse genetic study designs" (pdf on course homepage)

Define founder population and simulation parameters.

```
founders <- runMacs(nInd = 240,  
                   nChr = 10)  
  
simparam <- SimParam$new(founders)  
  
simparam$setSexes("yes_sys")  
  
## pedigree tracking  
simparam$setTrackPed(TRUE)
```

Define genetic architecture.

```
corA <- 0.5
simparam$addTraitA(nQtlPerChr = 10,
                   mean = c(0, 0),
                   var = c(1, 1),
                   corA = matrix(c(1, corA, corA, 1), ncol = 2))

simparam$setVarE(h2 = c(0.3, 0.3))
```


Build the breeding structure. 600 vs 600 animals.

```
pop <- newPop(founders,  
              simParam = simparam)  
  
offspring <- randCross(pop = pop,  
                      nProgeny = 10,  
                      nCrosses = 120,  
                      simParam = simparam)
```

Prepare simulated data.

```
environment <- rep(c(1, 2), 600)
```

```
pheno_data <- data.frame(animal = offspring@id,  
                          trait1 = scale(offspring@pheno[,1]),  
                          trait2 = scale(offspring@pheno[,2]),  
                          environment = environment,  
                          stringsAsFactors = FALSE)
```

```
pheno_data$trait1[environment == 2] <- NA
```

```
pheno_data$trait2[environment == 1] <- NA
```

Prepare simulated data.

```
ped <- data.frame(animal = c(pop@id, offspring@id),  
                  simparam$pedigree[, -3])  
ped$mother[ped$mother == 0] <- NA  
ped$father[ped$father == 0] <- NA
```

Run genetic analysis (MCMCglmm)

```
prior <- list(G = list(G1 = list(V = diag(2)/2, nu = 2)),  
             R = list(V = diag(2)/2, nu = 2))
```

```
model <- MCMCglmm(cbind(trait1, trait2) ~ 1,  
                 random = ~us(trait):animal,  
                 rcov = ~us(trait):units,  
                 family = c("gaussian", "gaussian"),  
                 pedigree = ped,  
                 data = pheno_data,  
                 prior = prior)
```

Summarise results. Heritability $h^2 = \frac{V_a}{V_a + V_e}$

```
h2_trait1 <- model$VCV[, "traittrait1:traittrait1.animal"] /  
  (model$VCV[, "traittrait1:traittrait1.units"] +  
    model$VCV[, "traittrait1:traittrait1.animal"])
```

```
h2_trait2 <- model$VCV[, "traittrait2:traittrait2.animal"] /  
  (model$VCV[, "traittrait2:traittrait2.units"] +  
    model$VCV[, "traittrait2:traittrait2.animal"])
```

Summarise results. Genetic correlation $r_g = \frac{\text{Cov}_{1,2}}{\sqrt{V_1 V_2}}$

```
rg <- model$VCV[, "traittrait1:traittrait2.animal"] /  
      sqrt(model$VCV[, "traittrait1:traittrait1.animal"] *  
            model$VCV[, "traittrait2:traittrait2.animal"])
```

Summarise results. Make a data frame.

```
data.frame(cor_a = corA,  
           cor_true = cor(pop@gv[,1], pop@gv[,2]),  
           rg_est = mean(rg),  
           rg_lower = HPDinterval(rg)[1],  
           rg_upper = HPDinterval(rg)[2],  
           h2_trait1 = mean(h2_trait1),  
           h2_trait1_lower = HPDinterval(h2_trait1)[1],  
           h2_trait1_upper = HPDinterval(h2_trait1)[2],  
           h2_trait2 = mean(h2_trait2),  
           h2_trait2_lower = HPDinterval(h2_trait2)[1],  
           h2_trait2_upper = HPDinterval(h2_trait2)[2])
```

Replicate. Make a grid of parameters, in this case correlations between additive effects (corresponding to genetic correlation).

```
cor_as <- rep(seq(from = -0.7, to = 0.7, by = 0.1), each = 50)
cor_as <- cor_as[!(cor_as >= -0.11 & cor_as < 0.11)]
```


Replicate. Wrap in functions.

```
results <- map_dfr(cor_as,  
                  function(x) {  
  
                      simulation <- make_simulation(x)  
  
                      data <- get_data(simulation$founders,  
                                       simulation$offspring,  
                                       simulation$simparam)  
  
                      model <- estimate_mcmcglmm(data$pheno_data,  
                                                  data$ped)  
  
                      get_results(model,  
                                  simulation$offspring,  
                                  x,  
                                  simulation$true_cor_g)  
                  })
```

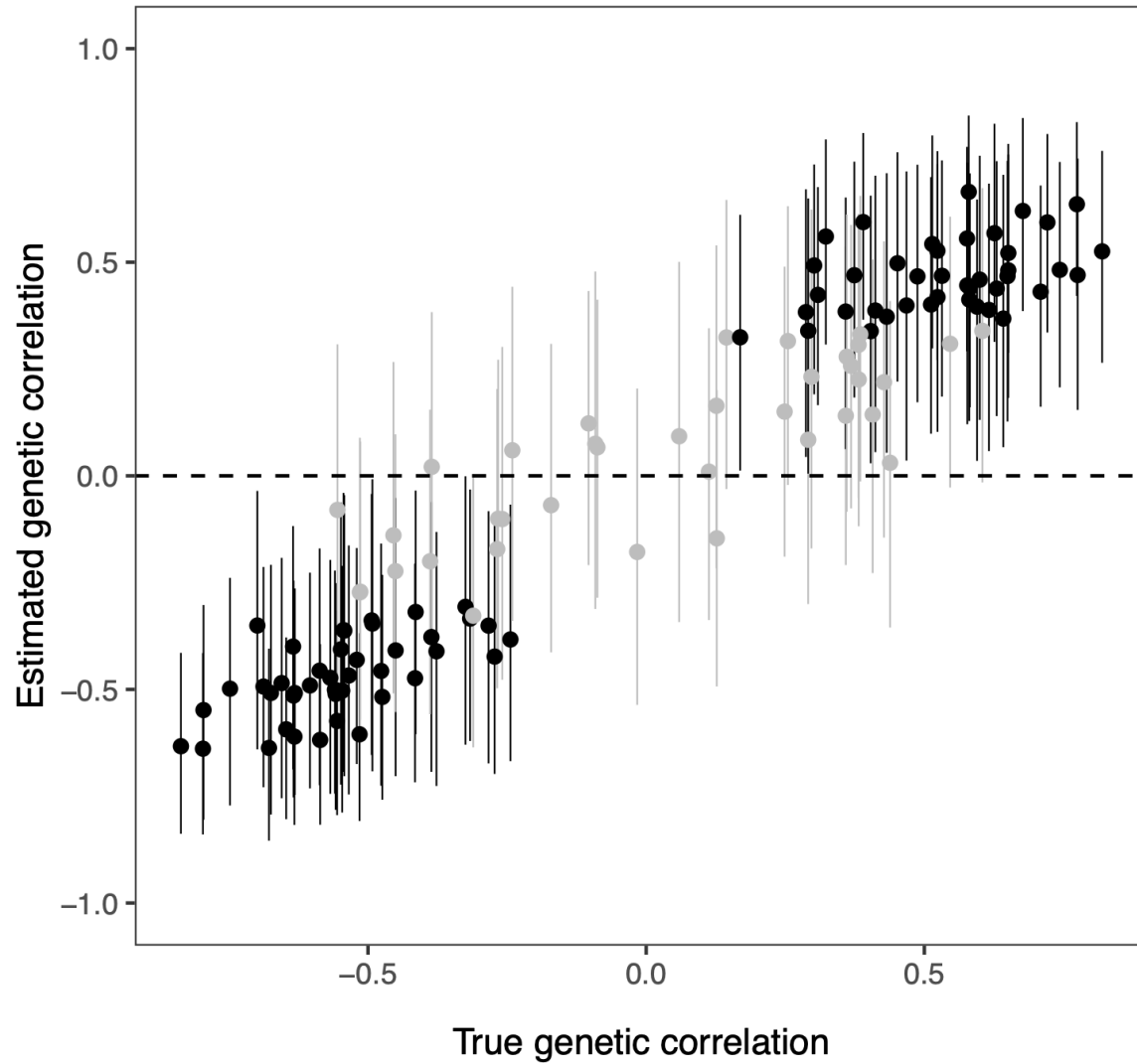
Visualise results.

```
results$significant <- ifelse(!(results$rg_lower <= 0 &  
                               results$rg_upper > 0),  
                              "significant",  
                              "not significant")
```

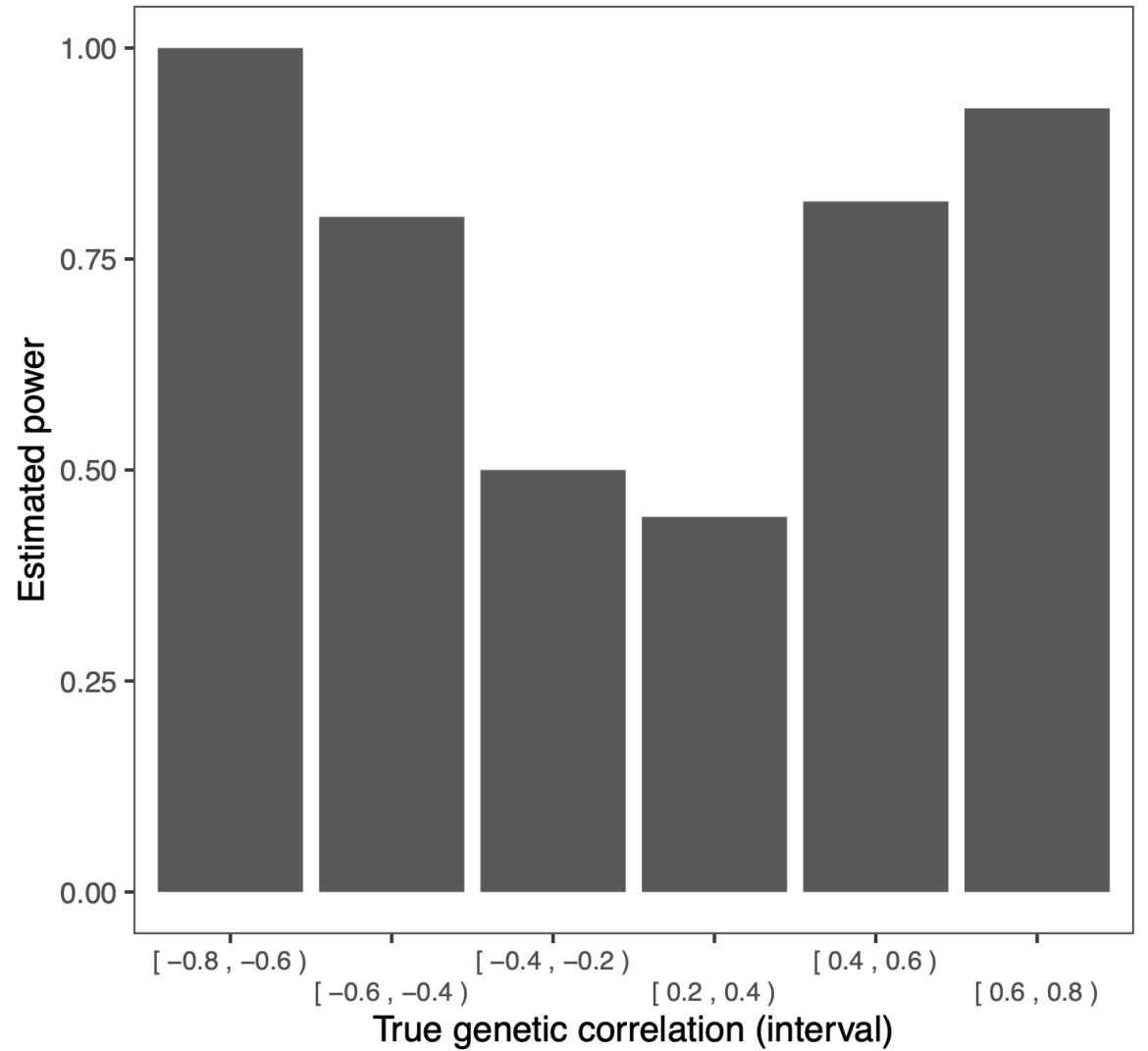
Visualise results.

```
plot_all_significant <- ggplot() +  
  geom_pointrange(aes(x = cor_true,  
                      y = rg_est,  
                      ymin = rg_lower,  
                      ymax = rg_upper,  
                      colour = significant),  
                 size = 0.25  
                 data = results) +  
  geom_hline(yintercept = 0,  
            linetype = 2) +  
  ylim(-1, 1) +  
  scale_colour_manual(values = c("grey", "black")) +  
  theme_bw() +  
  theme(panel.grid = element_blank()) +  
  xlab("True genetic correlation") +  
  ylab("Estimated genetic correlation") +  
  ggtitle("Estimates and credible intervals")
```

Estimates and credible intervals



Power



Controlling genetic effect sizes

The previous example was about genetic parameters. Easy to set with `addTraitAD`.

Sometimes we also need to control locus-level genetic coefficients with `importTrait` (e.g., power analysis of GWAS).

Example: assume we want one causative variant per chromosome, each with above 1% minor allele frequency.

One causative variant per chromosome, each with above 1% minor allele frequency.

```
n_chr <- founders_gwas@nChr
qtl_ix <- numeric(n_chr)
p <- numeric(n_chr)

threshold <- 0.01

for (chr in 1:n_chr) {
  geno <- pullSegSiteGeno(founders_gwas,
                           chr = chr,
                           simParam = simparam)

  f <- colSums(geno)/2/nrow(geno)

  candidate_qtl <- which(f > threshold & f < 1 - threshold)

  qtl_ix[chr] <- sample(candidate_qtl, 1)
  p[chr] <- f[qtl_ix[chr]]
}
```

The additive genetic variance due to a locus under random mating and linkage equilibrium (Falconer & Mackay 1996)

$$V_a = 2p(1 - p)[a + d(1 - 2p)]^2$$

$$\Rightarrow a = \sqrt{\frac{V'_a}{2p(1 - p)[1 + \delta(1 - 2p)]^2}}$$

Where V'_a is a desired additive genetic variance and we parameterise dominance with dominance degree $d = \delta|a|$

```
Va <- 0.1
delta <- 0.2
a <- sqrt(Va / (2 * p * (1 - p) * (1 + delta * (1 - p - p))^2))
d <- a * delta
```


Add the trait to the SimParam object:

```
trait <- new("TraitAD",  
            intercept = 0,  
            nLoci = as.integer(n_chr),  
            lociPerChr = rep(1L, n_chr),  
            lociLoc = as.integer(qtl_ix),  
            addEff = a,  
            domEff = d)  
  
simparam_gwas$manAddTrait(lociMap = trait,  
                           varE = 3)
```

Outline of simulation of genetic study designs

1. Define the founder population history.
2. Define simulation parameters.
3. Define genetic architecture.
4. Build the breeding structure.
5. Prepare simulated data.
6. Run genetic analysis.
7. Summarise results.
8. Repeat.
9. Visualise outcomes.

Power and design analysis are important and underappreciated, perhaps because they are hard work.

Simulation makes them less hard work.

Some simplifying assumptions are perfectly fine, but you probably need more data than you think.



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